



Tomas Monopoli

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Location: Aalborg, Denmark

[Website](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | [Web of Science](#)

Core Competences: *Physics-Informed AI, Bayesian Inference & Statistics, Uncertainty Quantification (UQ), Gaussian Process Regression (GPR/Kriging), Deep Learning (Transformers, CNNs), Power Electronics, Digital Twins, Predictive Maintenance, EMC and Space Systems.*

Technical Skills: *Coding* | Python (PyTorch, TensorFlow, Scikit-learn, numpy, pandas), MATLAB, LaTeX, C++ | *Simulation & Tools* | Ansys HFSS, COMSOL, SPICE, Git, Docker, Linux, VS Code

Career Summary [List of Publications]

PhD Engineer with over 6 years of experience in physical modeling and AI-driven system optimization. Bridging the gap between theoretical physics and industrial applications through robust machine learning pipelines and advanced statistical methods. Proven track record in leading high-impact research projects for energy and space sectors, by bridging the gap between theoretical data science and practical hardware engineering.

Experience

Postdoc in AI and Power Electronics | AAU, Aalborg, DK 2025 – Now

- Leading research for the [PHY-CALIPER Project](#) to discover "unknown physics" and calibrate predictive models for next-gen energy systems.
- Optimizing predictive maintenance workflows using **statistical ML**, **PIML**, and **Bayesian methods**.

Visiting Research Fellow | **European Space Agency (ESA)** / **ESTEC**, Noordwijk, NL 2021

- Developed data-driven **wideband EM modeling** for CubeSats, replacing heavy full-wave simulations with efficient equivalent models to **accelerate EMC testing** ([ESA project](#)).

R&D Engineer | **Edison**, Milan, IT 2019 – 2020

- Orchestrated R&D innovation projects, bridging technical feasibility with corporate business strategy.

Summer Internship | **CERN** ([CESP Program](#)), Geneva, CH 2019

Education

PhD in Electrical Engineering (cum Laude) | **Politecnico di Milano**, IT 2020 – 2025

- Joint Research with ESA/ESTEC:** Developed AI-driven optimization frameworks for modeling satellite electromagnetic environments, replacing physical testing with data-driven predictions.

Alta Scuola Politecnica (ASP) | **PoliMi & PoliTo**, Milan/Turin, IT 2018 – 2020

- Excellence Program:** Selected among the top 1% of students for a double-degree program focused on multidisciplinary innovation.

MSc in Electrical Engineering (110/110 cum Laude) | **PoliMi & PoliTo** 2017 – 2020

BSc in Electrical Engineering (110/110) | **PoliMi** 2014 – 2017

Technical Projects

AI-powered OCR for Scientific Papers | [Project Link](#) | [GitHub](#) 2024

- Developed a pipeline using **Transformer architectures** to accurately extract complex **LaTeX** expressions and structured data from PDFs. Optimized for **RAG (Retrieval-Augmented Generation)** workflows with LLMs.

SmartBin: Waste Classification System | Entrepreneurship Program 2018 – 2019

- Built a computer-vision-equipped bin using **CNNs** on Raspberry Pi; The prototype was awarded a presentation slot at **Italian Tech Week** in Turin.

Scientific Engagement

- Technical Reviewer** for scientific journals (TEMC, LECPA, IEEE Access) and AEES 2023 conference
- Presenter** at conferences (ESA workshop on Aerospace EMC 2025, COMPO 2024, GEMCCON 2020) and **Session Chair** for PTSET 2023